

Data Replication

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2025/10/10

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1 Definitions

- **LCE patents:** Low-carbon Energy patents, defined by the IEA report of IEA-Patents and the Energy Transition (2021). (see details in [Appendix A](#))
- **Invention:** An invention refers to a simple patent family.(see details about different types of patent families in [Appendix B](#))
- **TechField:** Each LCE invention is tagged with technology fields by CPC codes, with a TechField-subclass1-subclass2 format. There are three Techfields in total: End-use, Enabling and Energy supply. (see details in [Appendix A](#))
- **Priority patent:** A priority is defined as the patent with the earliest application date in a simple patent family. Accordingly, its application date is the **priority date**, and its patent office is the **priority authority**.^a
- **Inventor country:** The inventor country of an invention is defined as the inventor country of the priority patent.
- **Transfer:** A transfer is defined as an invention filed in a receiving country different from its priority authority, where the **receiving country** refers to the authorities of subsequent filings of the priority patent. For example, if a priority patent filed in Chinese patent office (priority authority) was later filed in Japanese patent office (receiving country), it is defined as a transfer from China to Japan. The transferred invention is called an **international patent family (IPF)**.

^aThis is a simplified definition adopted by previous literature (Dechezleprêtre et al. (2011) and Touboul et al. (2023)). In reality, one patent family may have several priority patents.

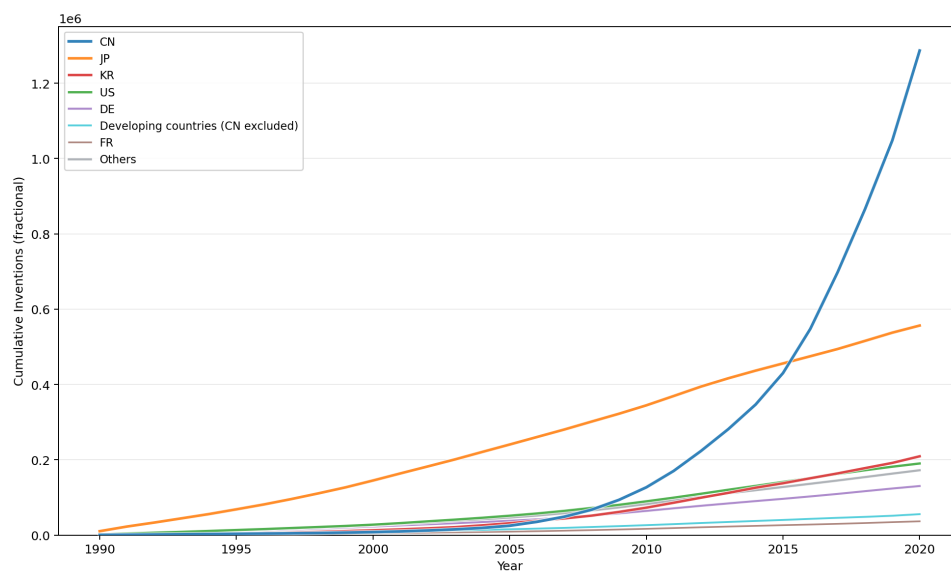
2 Data

- **Source:** PATSTAT, 2024 Autumn Version; updated by European Patent Office (EPO) biannually.
- **Coverage:** Across all technologies, 99,678,274 inventions in total; For LCE technologies, 4,889,113 LCE inventions. 96 authorities are involved in the selected sample of LCE patents.
- **Time Period:** From 1866-04-25 to 2024-06-04. By default, the time period in this report is from 1990-01-01 to 2020-12-31 for better data quality. ¹ In the analysis of transfers (section 4), I focus on 1900-01-01 to 2022-12-31 to show the dynamics of transfer patterns over a longer period.
- **Data issue:** Most of the information about inventor location is missing. I use authority countries as substitutes for missing inventor countries, which is adopted by previous literature(Dechezleprêtre et al. (2011),Touboul et al. (2023)), as well as the report of IEA-Patents and the Energy Transition (2021); I also construct an adjusted method to fill in missing information of priorities with information from their subsequent filings (see details in [Appendix C](#)).

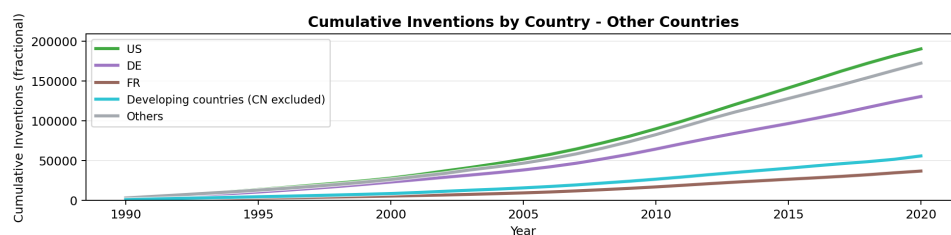
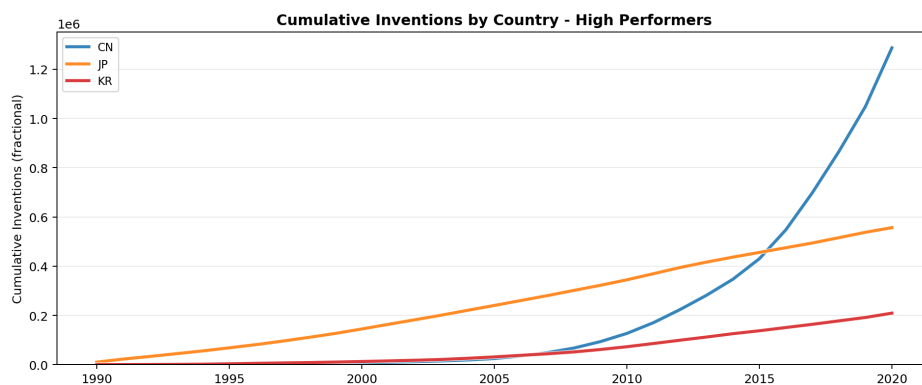
¹Many developing countries haven't built patent offices or standardized data records until late in the 20th century. For example, the Chinese patent office was established in 1985, and its data became available only afterwards. For the most recent data (2020-), the formation of a patent family may not be complete, with many of the patent families being singletons since patent applications typically take years.

3 Trends

3.1 By inventor country



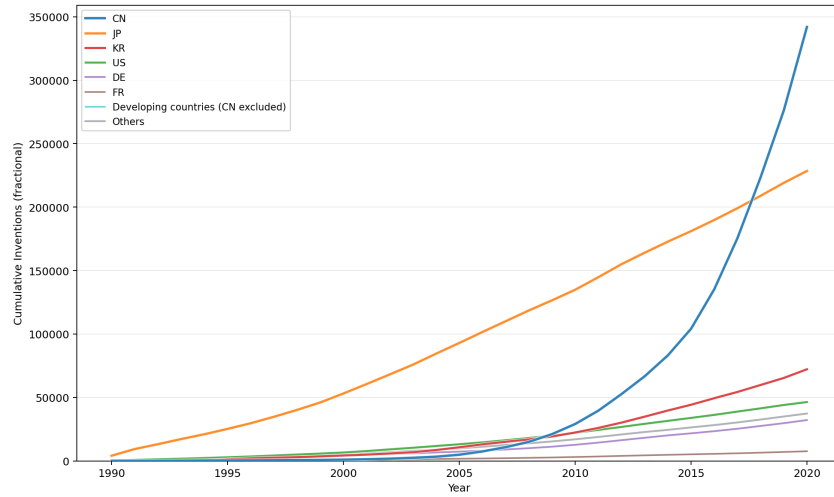
(a)



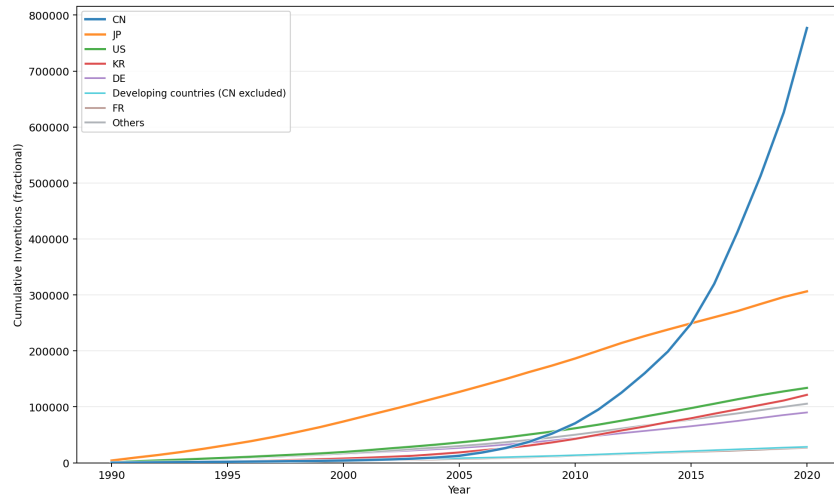
(b)

Figure 1: Cumulative inventions by inventor country, 1990-2020

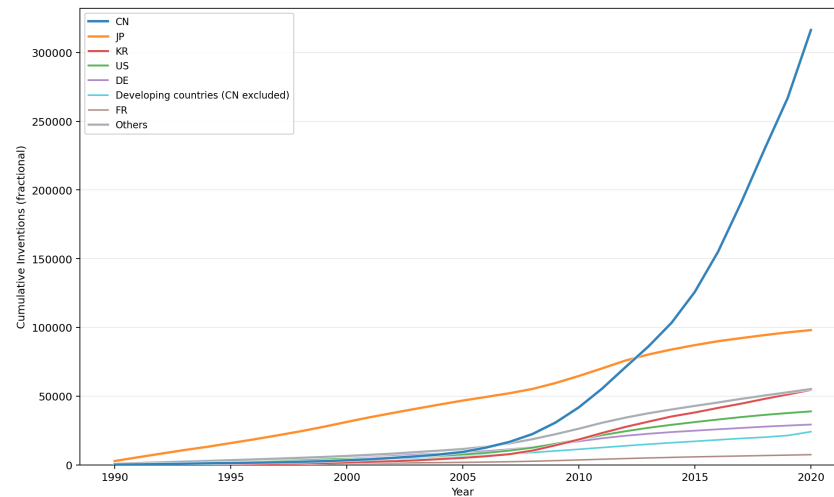
Notes: These figures plot cumulative inventions by inventor country from 1990–2020. Figure (a) shows all countries on a common scale; figure (b) with dual scale separates “high performers” (CN, JP, KR) from “other countries” (threshold set at 15% of the maximum cumulative inventions). An invention refers to a simple patent family, weighted equally across all inventor countries. For example, if there are two inventor countries of an invention, then each country is counted as one-half. The invention year is the priority year of that invention. Cumulatives start at each country’s first observed year in the dataset.



(a) Enabling technologies



(b) End-use technologies

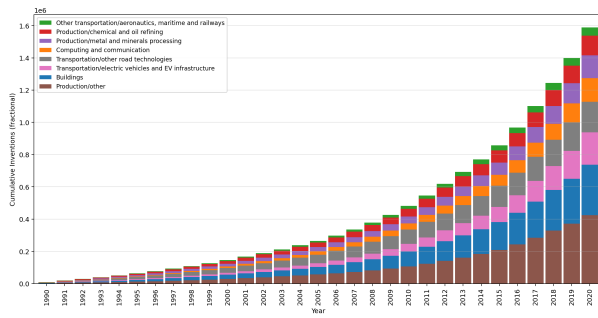


(c) Energy supply technologies

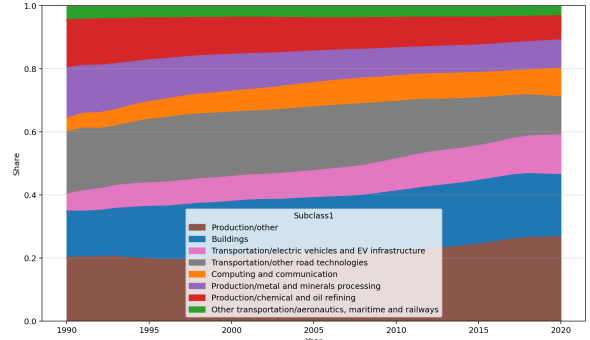
Figure 2: Cumulative inventions by technology fields, 1990–2020.

Notes: These figures plot cumulative inventions by inventor country from 1990–2020 for (a) enabling technologies, (b) end-use technologies, and (c) energy supply technologies. Country series are shown for CN (mainland China), JP, US, KR, DE, FR, an aggregate of “Developing countries (CN excluded)” and a residual “Others”. For each invention, weight 1 is split equally across all inventor countries. The invention year is the priority year of that invention. Cumulatives start at each country’s first observed year in the dataset.

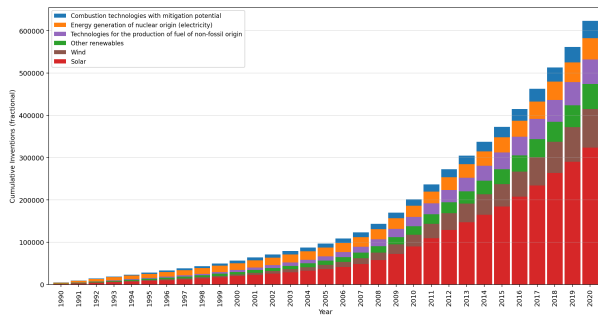
3.2 By Techfields



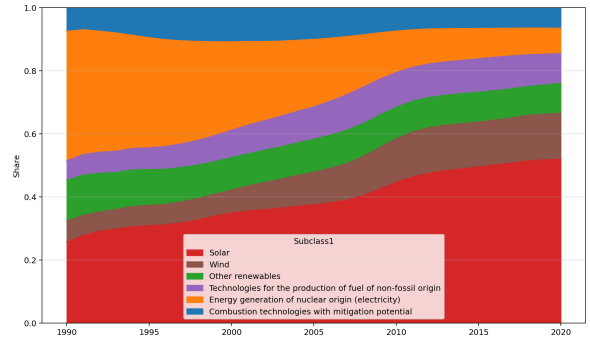
(a) End-use: cumulative totals



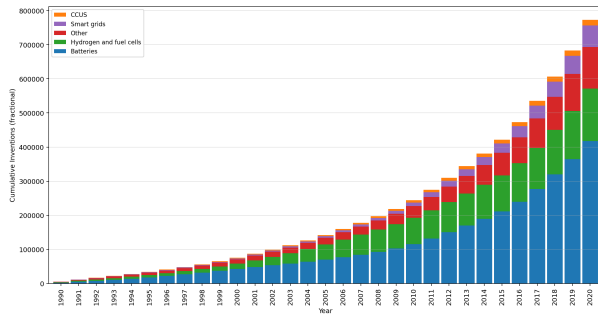
(b) End-use: cumulative shares



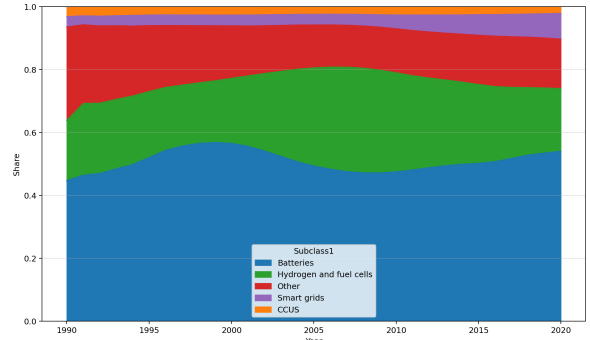
(c) Energy supply: cumulative totals



(d) Energy supply: cumulative shares



(e) Enabling: cumulative totals



(f) Enabling: cumulative shares

Figure 3: Subclass1 Composition in Technology Fields, 1990–2020.

Notes: These figures present the cumulative inventions of each Techfields (End-use Technologies, Low-carbon energy supply, and Enabling Technologies), showing both the cumulative totals (left panels) and shares of each Subclass1 within the TechField (right panels). Each LCE invention is tagged with technology fields by CPC codes, with a TechField-subclass1-subclass2 format (see section A). Units are fractional inventions by priority year; for any invention that maps to multiple Subclass1 entries within a given TechField, the weight is split equally across those Subclass1. For example, if an invention of enabling technologies maps to both CCUS and batteries, it's counted as one-half in each subclass1. Cumulative totals start at the first observed year of the invention and are summed through 2020. Cumulative shares are the share of each subclasses' cumulative totals in that year.

4 Transfer

4.1 Transfer Rates

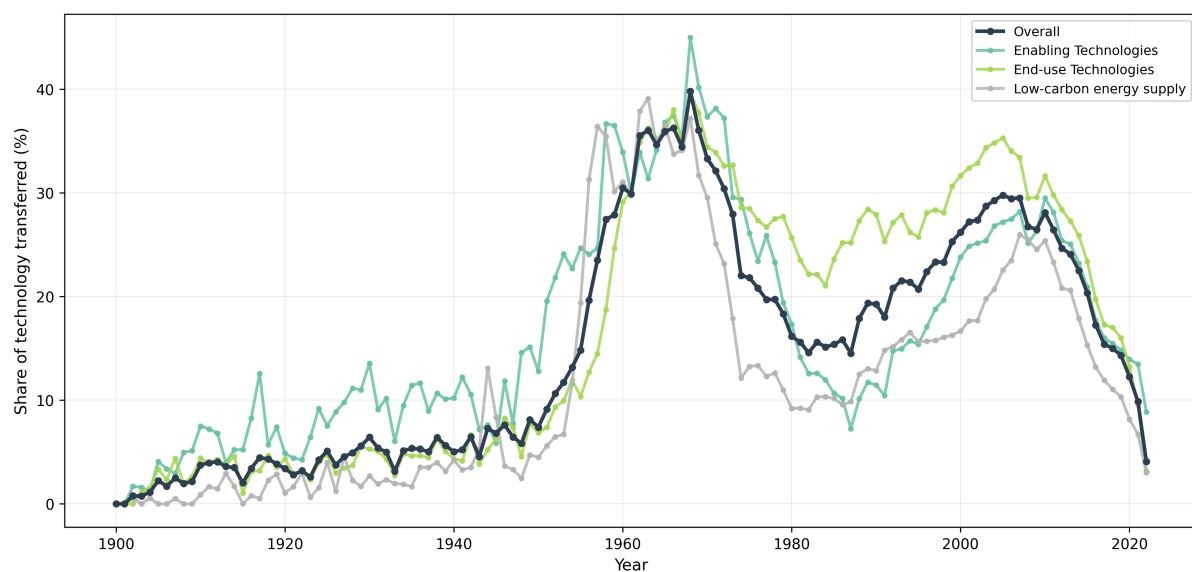
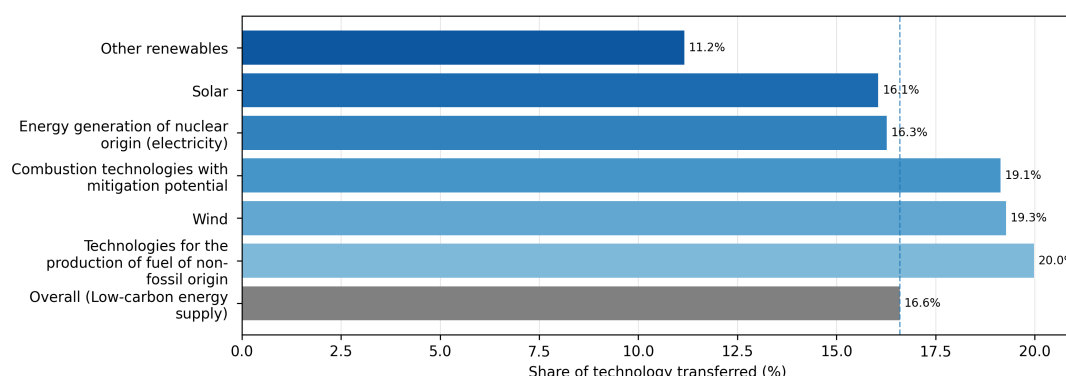


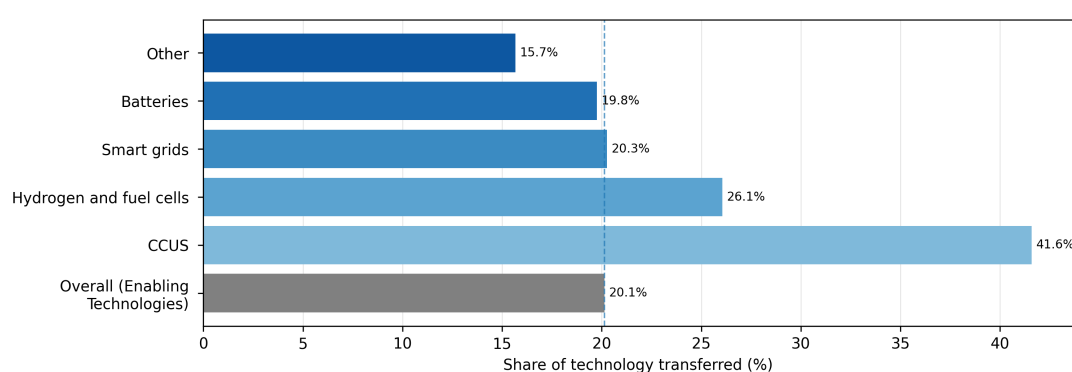
Figure 4: Transfer Rates of LCE Inventions, 1900-2022

Notes: This figure plots annual transfer rates of LCE inventions from 1900 to 2022, shown for both overall sample and three TechFields separately. The transfer rate in year t is defined as the annual share of international patent families (IPF, i.e., inventions filed at least one application authority different from its priority authority) in all inventions. Each TechField series is computed within its field-year pair; if an invention maps to multiple fields, it contributes once to every relevant series without weighting.

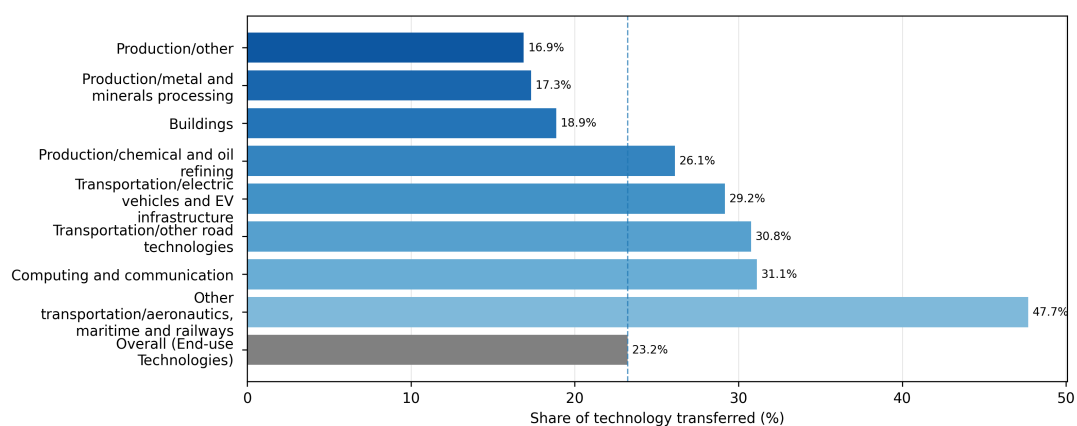
4.2 Transfer rates by subclass1



(a) Energy supply technologies



(b) Enabling technologies



(c) End-use technologies

Figure 5: Transfer rates by Subclass1, 1990–2020.

Notes: The figure reports subclass1-level transfer rates for LCE inventions between 1990 and 2020, shown separately for (a) energy-supply, (b) enabling, and (c) end-use technologies. Each LCE invention is tagged with technology fields by CPC codes, with a TechField-subclass1-subclass2 format(see details in section A). Subclass1-level transfer rates are computed with the share of transfers in total inventions of that subclass. Inventions that map to multiple subclasses are retained in each subclass without weighting. The gray bar and vertical dashed line in each panel mark the corresponding TechField-level transfer rates as benchmark.

4.3 Transfer Distribution between Income Groups

(a) 1990–1995 (percent)				(b) 1995–2000 (percent)			
Origin \ Destination	High	Middle	Low	Origin \ Destination	High	Middle	Low
High	84.3 (85.7)	13.1 (13.0)	0.1 (0.1)	High	80.1 (81.8)	17.4 (16.7)	0.1 (0.1)
Middle	2.1 (1.1)	0.4 (0.2)	0.0 (0.0)	Middle	1.9 (1.2)	0.5 (0.3)	0.0 (0.0)
Low	0.0 (0.0)	0.0 (0.0)	0.0 (0.0)	Low	0.0 (0.0)	0.0 (0.0)	0.0 (0.0)
(c) 2000–2005 (percent)				(d) 2005–2010 (percent)			
Origin \ Destination	High	Middle	Low	Origin \ Destination	High	Middle	Low
High	74.9 (76.4)	21.1 (21.5)	0.1 (0.0)	High	65.7 (68.0)	28.2 (27.8)	0.1 (0.1)
Middle	3.2 (1.6)	0.7 (0.4)	0.0 (0.0)	Middle	4.9 (3.3)	1.1 (0.8)	0.0 (0.0)
Low	0.0 (0.0)	0.0 (0.0)	0.0 (0.0)	Low	0.0 (0.0)	0.0 (0.0)	0.0 (0.0)
(e) 2010–2015 (percent)				(f) 2015–2020 (percent)			
Origin \ Destination	High	Middle	Low	Origin \ Destination	High	Middle	Low
High	61.8 (63.2)	30.2 (29.8)	0.1 (0.1)	High	58.8 (58.4)	27.7 (27.4)	0.0 (0.0)
Middle	6.0 (5.4)	1.9 (1.5)	0.0 (0.0)	Middle	11.4 (12.5)	2.0 (1.7)	0.0 (0.0)
Low	0.0 (0.0)	0.0 (0.0)	0.0 (0.0)	Low	0.0 (0.0)	0.0 (0.0)	0.0 (0.0)

Table 1: Transfer distribution of LCE inventions between country income groups.

Notes: This table reports the distribution of transfers of LCE inventions across country income groups (High, Middle, and Low) in six five-year windows (from 1990 to 2020). Countries are mapped to High, Middle, and Low income groups using World Bank classifications, following Touboul et al. (2023) (see details in section D). The origin country refers to inventor countries of the priority patents, and the destination country refers to the authorities of subsequent filings of that priority patent. For each transfer, I first construct inventor-country $\times$ receiving-country pairs, and then weight by $1/k$ where k is the number of inventor countries, excluding self flows (i.e., same origin and destination country). For example, if the inventor countries are China and Korea, and the receiving countries are China and Mexico, I count this as a 1/2 China-Mexico transfer, a 1/2 Korea-Mexico transfer, and a 1/2 Korea-China transfer. Cell entries are then computed as the percentage shares of total weighted pairs of origin-destination transfer in the given time period. Values shown are baseline shares using authorities as substitutes for missing inventor countries, with percentages in parentheses recomputed using an adjusted method using inventor countries of subsequent filings as substitutes (see Appendix C).

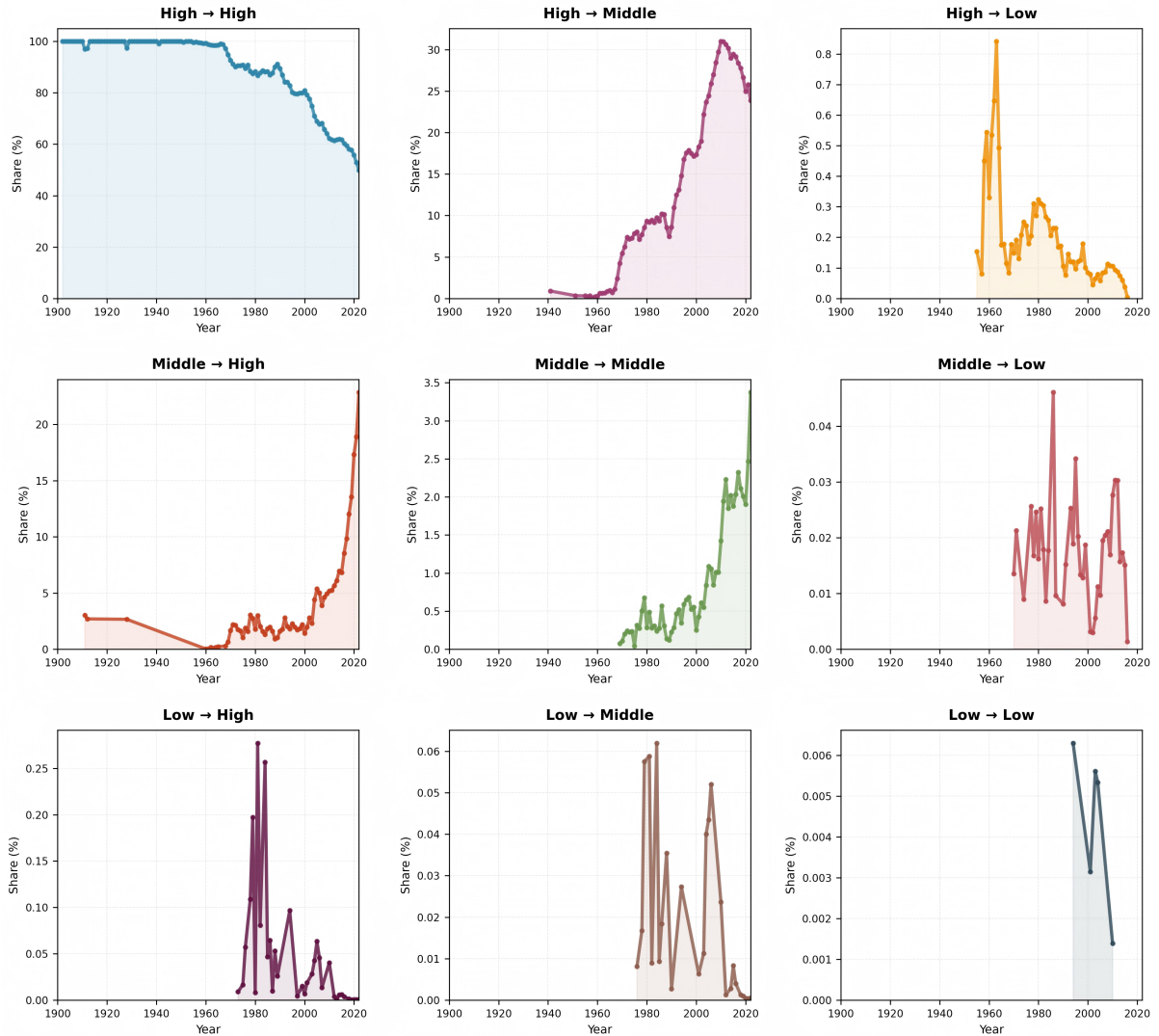
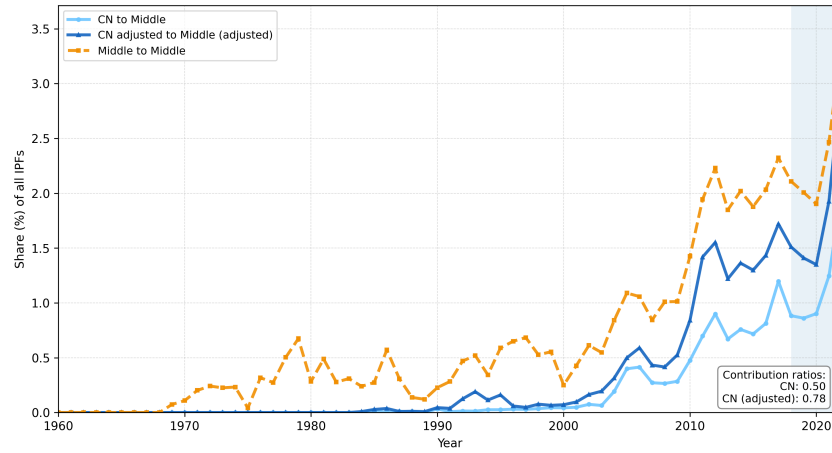


Figure 6: Time Trends of Technology Transfer Shares by Income Group, 1900–2022

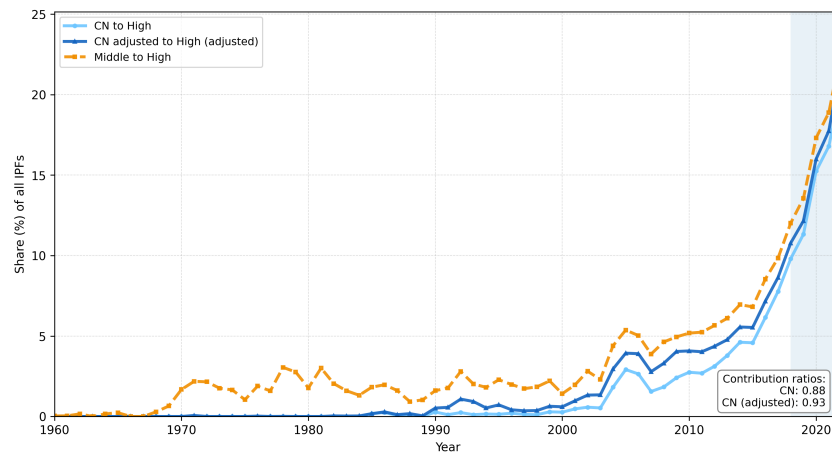
Notes: The figure extends table 1 to a dynamic time series panel of transfer distribution between country income groups. Each panel plots the annual share of transfers for a given origin–destination income-group pair, from 1900 to 2022. Country income group classification and share computation follow the same method of table 1.

4.4 China's outbound transfer

China's outbound transfer is the driving force of the middle-high, and middle-middle transfer increases.



(a) China's Contribution to Middle-High Transfer Increases



(b) China's Contribution to Middle-middle Transfer Increases

Figure 7: China's Contribution to Middle-high, Middle-middle transfer increases, 1960-2022

Notes: This figure plots China's contribution to increases in Middle-High and Middle-Middle transfers, over 1960 to 2022. Panel (a) shows annual shares for CN-High transfer versus Middle-High transfer; Panel (b) shows annual shares for CN-Middle transfer versus Middle-Middle transfer. The "CN adjusted" series aggregates CN (mainland, China) with HK (Hong Kong, China), MO (Macao, China), and TW (Taiwan, China). The shaded band marks the last five years of the window (2018–2022) used to compute the inset *contribution ratios*, which is defined as: the change in the CN (or CN adjusted) share between the first five years (1960–1964) and the last five years (2018–2022) divided by the corresponding change in the benchmark series. All shares use the same denominator: total transfers in that year. The computation of transfer weights follows the same method in section 4.3.

However, the increases seem to be driven by China's increasing share in global IPFs, not by structural changes of China's transfer patterns, which is relatively stable over time

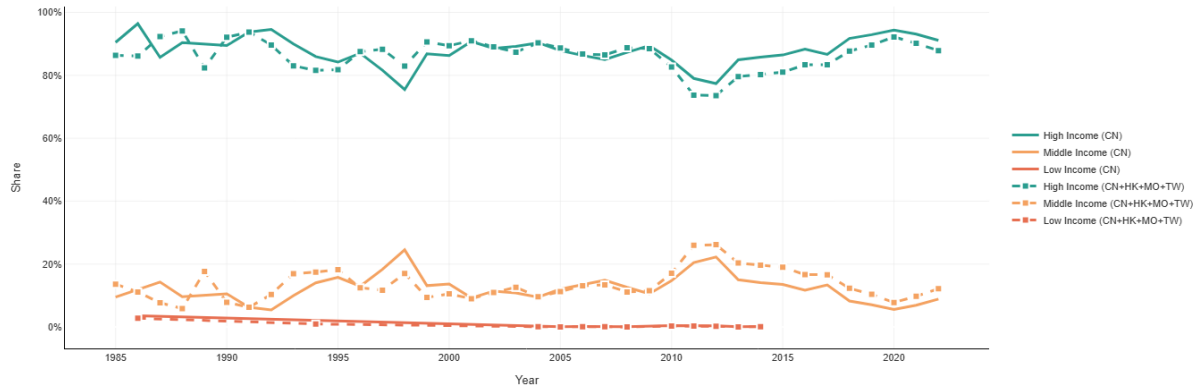


Figure 8: Shares of Destination Income Groups in China's Outbound Transfers, 1985-2022

Notes: This figure shows the annual composition of China's outbound transfers by destination income group (High, Middle, and Low), from 1985 to 2022, with solid lines for CN and dashed lines for "CN adjusted". Series plot the share of China-origin (or CN-adjusted-origin) transfer weight going to each income group, i.e., destination weight divided by the China's total outbound transfer weight that year. The computation of transfer weights follows the same method in section 4.3.

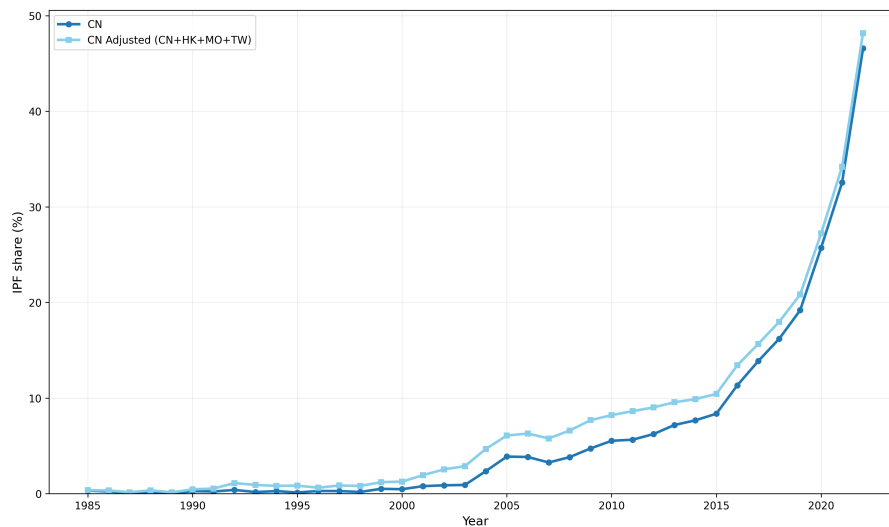


Figure 9: China's Global Share of International Patent Families, 1985-2022

Notes: This figure plots the global share of China's international patent families (IPF, or transfers), from 1985 to 2022. The denominator is IPFs in all countries. The computation of transfer weights follows the same method in section 4.3.

Destination countries of China's outbound transfer, 1985-2022

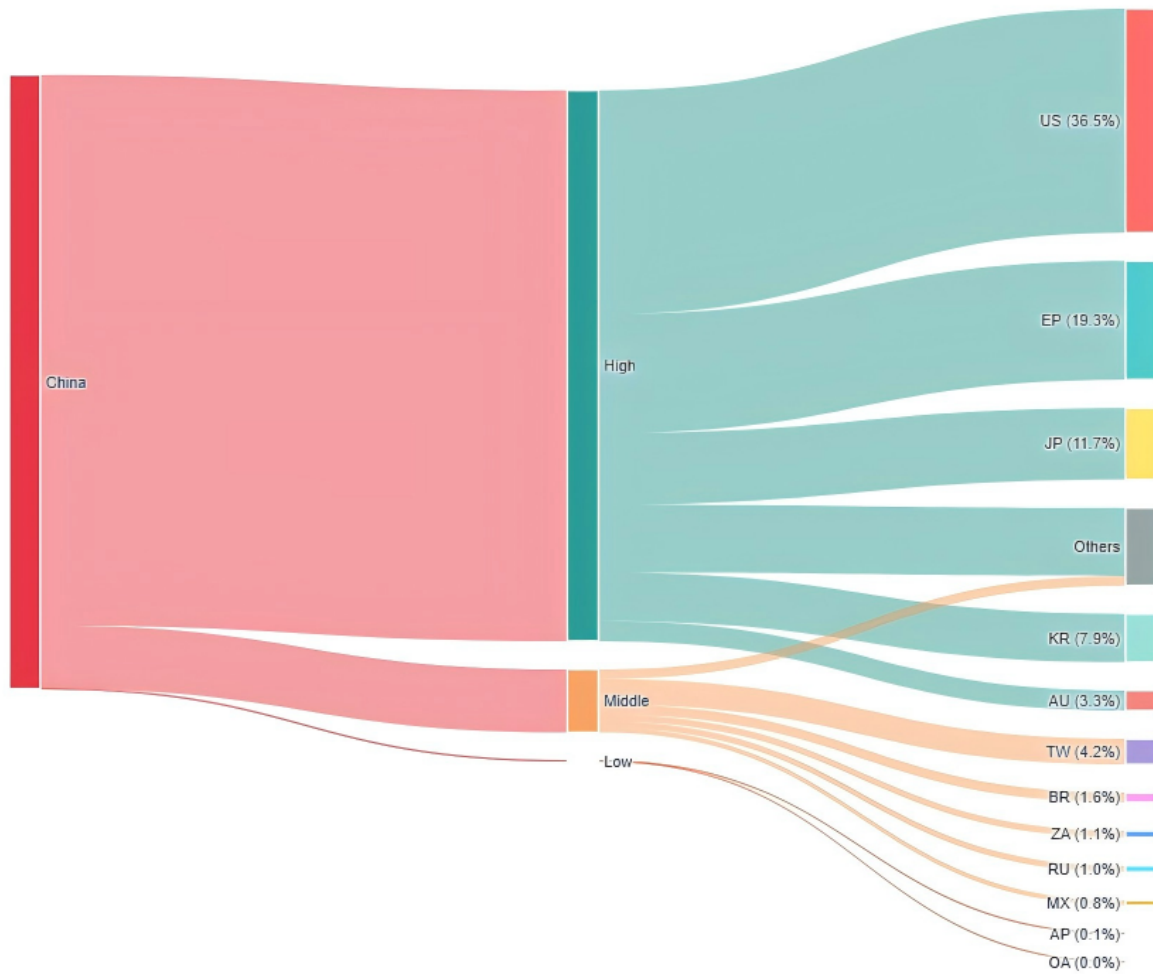


Figure 10: Destinations of China's Outbound Transfer

Notes: This Sankey diagram summarizes China's outbound transfer destinations. Each destination's share is computed as the CN-to-destination transfer weight divided by China's total outbound transfer over 1985 to 2022. Destinations are grouped by income class; within each class the top five recipient economies are displayed, with the remainder shown as "Others". The computation of transfer weight follows the same method in section 4.3. Note that I exclude "WO" for destinations.²

²"WO" is an intermediate organization for PCT patents to later enter regional/national patent offices, and is not a destination jurisdiction. Specifically, each PCT application grants a priority right in over 150 contracting countries of WIPO; and applicants still need to file in regional/national authorities (e.g., JP, US, EP) in later phases. When PCT patents enter receiving countries, they would get a new application number and the authority country record. Therefore, the receiving countries would be included as destinations, and there's no need to include WO. If PCT patents don't end up entering any regions/countries, WO should also be excluded, since this would not be considered a valid transfer

A LCE Patents

Source: [Patents and the Energy Transition –Analysis - IEA](#)

Notes: the [CPC codes](#) involved cover Y04S, and all Y02* except for Y02A (adaptation technologies) and Y02W (mitigation technologies related to wastewater treatment or waste management).

Table 2

TechField	Subclass1	Subclass2	CPC code
Low-carbon energy supply	Wind	—	Y02E10/70 OR Y02E10/72 OR Y02E10/727 OR Y02E10/728 OR Y02E10/74 OR Y02E10/76
Low-carbon energy supply	Solar	Solar PV	Y02E10/50 OR Y02E10/52 OR Y02E10/541 OR Y02E10/542 OR Y02E10/543 OR Y02E10/544 OR Y02E10/545 OR Y02E10/546 OR Y02E10/547 OR Y02E10/548 OR Y02E10/549 OR Y02E10/56
Low-carbon energy supply	Solar	Solar thermal	Y02E10/40 OR Y02E10/44 OR Y02E10/46 OR Y02E10/47
Low-carbon energy supply	Solar	Other solar	Y02E10/60
Low-carbon energy supply	Other renewables	Geothermal energy	Y02E10/10
Low-carbon energy supply	Other renewables	Hydro	Y02E10/20
Low-carbon energy supply	Other renewables	Marine	Y02E10/30
Low-carbon energy supply	Other renewables	Other	Y02E10/00
Low-carbon energy supply	Technologies for the production of fuel of non-fossil origin	Biofuels	Y02E50/10
Low-carbon energy supply	Technologies for the production of fuel of non-fossil origin	Fuel from waste	Y02E50/30
Low-carbon energy supply	Technologies for the production of fuel of non-fossil origin	Other	Y02E50/00
Low-carbon energy supply	Combustion technologies with mitigation potential	—	Y02E20/00 OR Y02E20/12 OR Y02E20/14 OR Y02E20/16 OR Y02E20/18 OR Y02E20/30 OR Y02E20/32 OR Y02E20/34
Low-carbon energy supply	Energy generation of nuclear origin (electricity)	—	Y02E30/00 OR Y02E30/10 OR Y02E30/30
Enabling Technologies	CCUS	—	Y02C20/00 OR Y02C20/10 OR Y02C20/20 OR Y02C20/30 OR Y02C20/40

Continued on next page

TechField	Subclass1	Subclass2	CPC code
Enabling Technologies	Batteries	—	Y02E60/10
Enabling Technologies	Hydrogen and fuel cells	—	Y02E60/30 OR Y02E60/32 OR Y02E60/34 OR Y02E60/36 OR Y02E60/50
Enabling Technologies	Other	—	Y02E60/00 OR Y02E60/13 OR Y02E60/14 OR Y02E60/16 OR (Y02E70/00 OR Y02E70/30) OR Y02E60/60 OR Y02E40/00 OR Y02E40/10 OR Y02E40/20 OR Y02E40/30 OR Y02E40/40 OR Y02E40/50 OR Y02E40/60
Enabling Technologies	Smart grids	—	Y04S
End-use Technologies	Buildings	—	Y02B
End-use Technologies	Production/chemical and oil refining	—	Y02P20/00 OR Y02P20/10 OR Y02P20/129 OR Y02P20/133 OR Y02P20/141 OR Y02P20/143 OR Y02P20/145 OR Y02P20/151 OR Y02P20/155 OR Y02P20/156 OR Y02P20/20 OR Y02P20/30 OR Y02P20/40 OR Y02P20/50 OR Y02P20/52 OR Y02P20/54 OR Y02P20/55 OR Y02P20/582 OR Y02P20/584 OR Y02P20/59 OR Y02P30/00 OR Y02P30/20 OR Y02P30/40
End-use Technologies	Production/metal and minerals processing	—	Y02P10/00 OR Y02P10/10 OR Y02P10/122 OR Y02P10/134 OR Y02P10/143 OR Y02P10/146 OR Y02P10/20 OR Y02P10/25 OR Y02P10/32 OR Y02P40/00 OR Y02P40/10 OR Y02P40/121 OR Y02P40/125 OR Y02P40/18 OR Y02P40/40 OR Y02P40/45 OR Y02P40/50 OR Y02P40/57 OR Y02P40/60
End-use Technologies	Production/other	Agriculture	Y02P60/00 OR Y02P60/12 OR Y02P60/14 OR Y02P60/20 OR Y02P60/21 OR Y02P60/22 OR Y02P60/30 OR Y02P60/40 OR Y02P60/50 OR Y02P60/52 OR Y02P60/60 OR Y02P60/80 OR Y02P60/85 OR Y02P60/87
End-use Technologies	Production/other	Consumer products	Y02P70/00 OR Y02P70/10 OR Y02P70/50 OR Y02P70/62

Continued on next page

TechField	Subclass1	Subclass2	CPC code
End-use Technologies	Production/other	Other production	Y02P80/00 OR Y02P80/10 OR Y02P80/14 OR Y02P80/15 OR Y02P80/20 OR Y02P80/30 OR Y02P80/40 OR Y02P90/00 OR Y02P90/02 OR Y02P90/30 OR Y02P90/40 OR Y02P90/45 OR Y02P90/50 OR Y02P90/60 OR Y02P90/70 OR Y02P90/80 OR Y02P90/82 OR Y02P90/84 OR Y02P90/845 OR Y02P90/90 OR Y02P90/95
End-use Technologies	Transportation/electric vehicles and EV infrastructure	EV and infrastructure	Y02T10/60 OR Y02T10/62 OR Y02T10/64 OR Y02T10/70 OR Y02T10/7072 OR Y02T10/72 OR Y02T10/92 OR Y02T90/10 OR Y02T90/12 OR Y02T90/14 OR Y02T90/16 OR Y02T90/167
End-use Technologies	Transportation/electric vehicles and EV infrastructure	Fuel cells for road vehicles	Y02T90/40
End-use Technologies	Transportation/other road technologies	—	Y02T10/00 OR Y02T10/10 OR Y02T10/12 OR Y02T10/30 OR Y02T10/40 OR Y02T10/80 OR Y02T10/82 OR Y02T10/84 OR Y02T10/86 OR Y02T10/88 OR Y02T10/90 OR Y02T90/00
End-use Technologies	Other transportation/aeronautics, maritime and railways	Aeronautics	Y02T50/00 OR Y02T50/10 OR Y02T50/30 OR Y02T50/40 OR Y02T50/50 OR Y02T50/60 OR Y02T50/678 OR Y02T50/80
End-use Technologies	Other transportation/aeronautics, maritime and railways	Maritime and waterways	Y02T70/00 OR Y02T70/10 OR Y02T70/50 OR Y02T70/5218 OR Y02T70/5236
End-use Technologies	Other transportation/aeronautics, maritime and railways	Railways	Y02T30/00
End-use Technologies	Computing and communication	—	Y02D10/00 OR Y02D30/00 OR Y02D30/50 OR Y02D30/70

B Patent Families

PATSTAT has two types of patent family, one is simple patent family, defined as a group of patents sharing exactly the same set of patent priorities; and the other is INPADOC extended family, defined as a group of patents having at least one priority in common with at least one of the other members in the family. For example, in the figure below, we have four simple families (marked with family1-4), but only one INPADOC family.

Document 1	Priority P1			Family 1
Document 2	Priority P1	Priority P2		Family 2
Document 3	Priority P1	Priority P2		Family 2
Document 4		Priority P2	Priority P3	Family 3
Document 5			Priority P3	Family 4

The PATSTAT has more standardized data for simple patent families, including harmonized CPC codes of each simple family, as well as citations between simple families. Therefore, this report starts with focusing on simple family, following previous literature and reports (Dechezleprêtre et al. (2011), Touboul et al. (2023), IEA-Patents and the Energy Transition (2021)).

In extended patent families, PATSNAP patent family has better data quality. Since not all of the information (i.e. priority number, application number) are normalized, it will lead to some patents missing from families and being represented as individual patents with no family. PATSNAP deals with this problem and develops the PATSNAP patent family following the same rules as INPADOC, but by a more powerful normalization of patent numbers, so that patents can be grouped into families correctly. For example, if the priority number format from the New Zealand patent office adds a couple of zeroes in the middle of the number (NZ00123), but the Japanese patent office does not (JP123), PATSNAP would remove the zeroes so that the priority numbers match and the patents can be grouped.

C Missing Inventors' Countries

Statistics:

INVENTOR_RESIDENCE_CTRY: 4,742,726 rows (97.01%) missing
APPLICANT_RESIDENCE_CTRY: 4,705,266 rows (96.24%) missing
APPLICANT_CTRY: 3,865,838 rows (79.07%) missing
INVENTOR_CTRY: 3,986,677 rows (81.54%) missing

A large proportion of country information of LCE inventions is missing, shown in the box above. Dechezleprêtre et al. (2011) provides two solutions: (i) assuming **priority office** (the patent office of the priority patent) to be inventor's country, because of the "home-bias" (i.e., most priority applications are filed by local inventors); (ii) filling in missing countries based on the **average proportion of available inventor countries for the same technology field filed in the same patent office**. However, method (ii) is not biased only if the missing inventors are randomly drawn from the overall inventors' pool of the same tech field, in the same patent office. Since most of available inventor countries are the application authority itself, these two methods don't have a big difference, and Dechezleprêtre et al. (2011) adopts the first method as baseline. Following the paper, this report also adopts method (i) as a baseline. I conducted several validation tests, and the replicated results closely match the findings reported in previous literature.

As a more careful adjustment, I construct method (iii) following de Rassenfosse and Seliger (2021). Since the target is inventor countries of priority patents, I leverage **information from subsequent filings that claim that priority to backfill missing inventor countries**. The imputation proceeds in a fixed order: (1) use the inventor country on the priority itself if available; (2) otherwise, consult subsequent filings claiming that priority, with a preference for the direct equivalent (a subsequent application that claim the target priority as its only priority³) and then other subsequent filings; (3) if inventor country remains missing, substitute the applicant country, applying the same source order (first filing → direct equivalents → other subsequent filings); (4) if all are missing, use the authority (office) of the first filing as a last resort. See flowchart fig. 11 in next page.

³Sometimes, a patent document in office A can give rise to two or more patent documents in secondary office B (case of split applications). Conversely, two or more patent documents in office A can be merged into one application at office B.

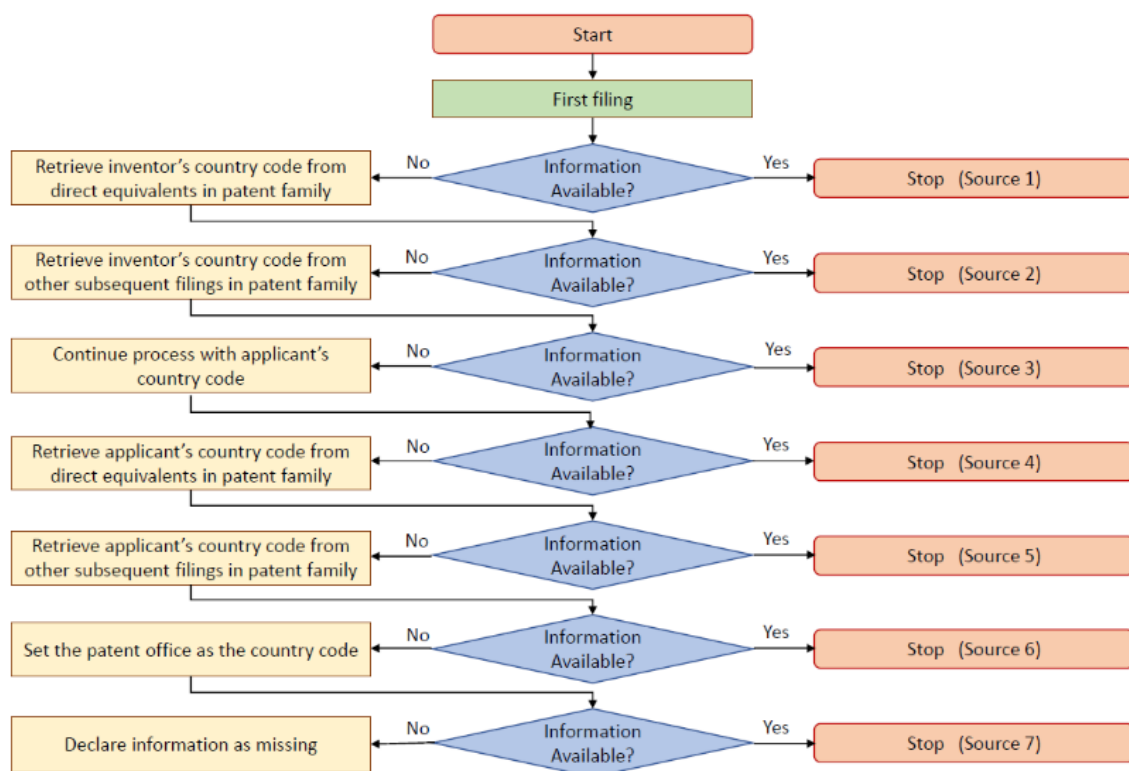


Figure 11: The flowchart of the imputation algorithm for inventor country code

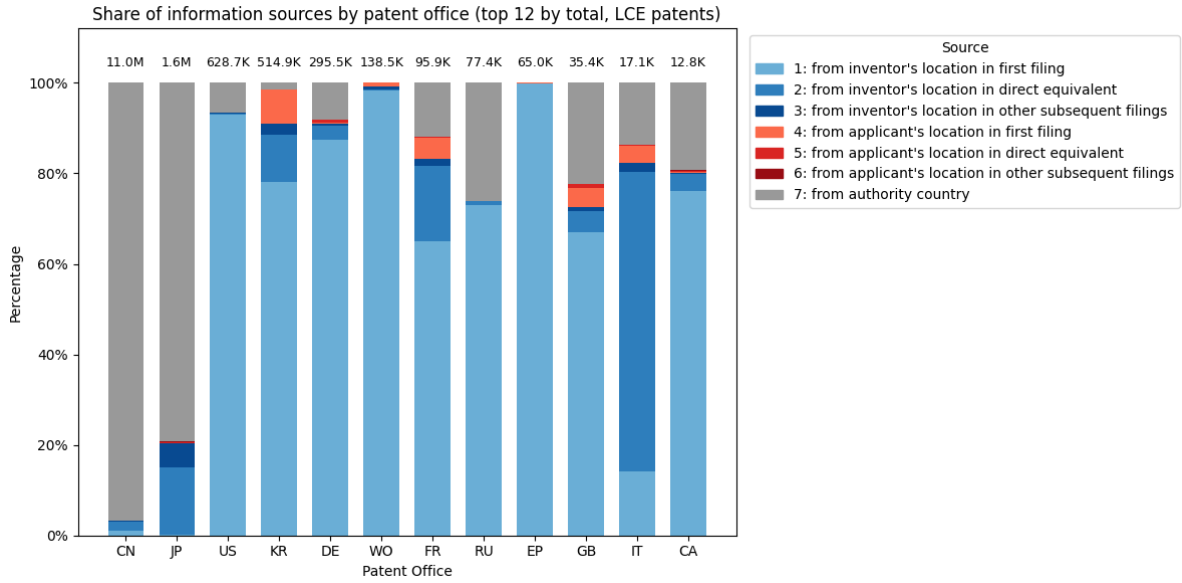


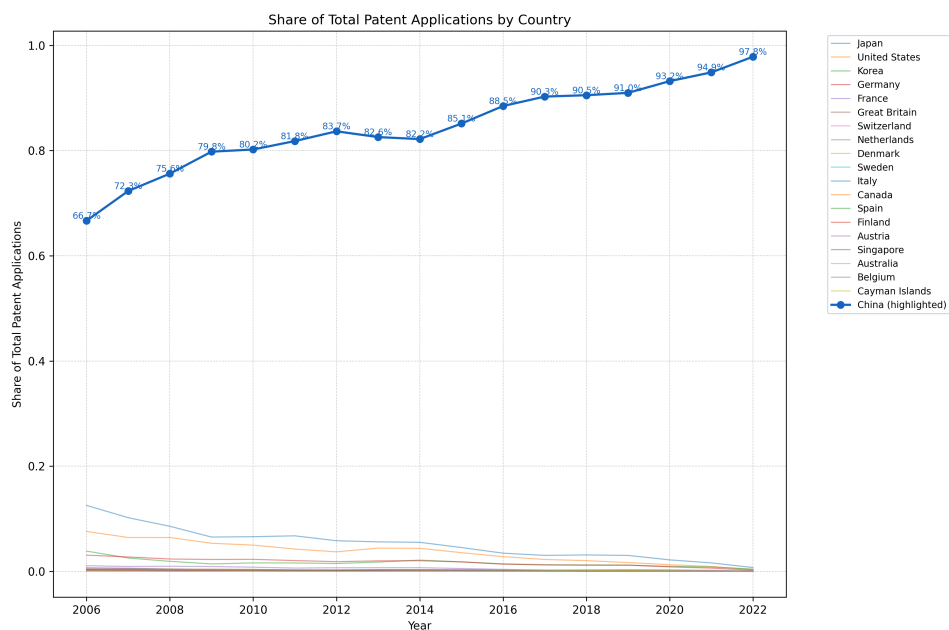
Figure 12: Sources of inventor-country information by office

The result of method (iii) is shown in fig. 12. Most of the missing inventor's countries come from two patent offices: CN and JP. The large share of source 7 substitute in these two countries is partly due to the data processing rule of PATSTAT. In PATSTAT, Table TLS207_PERS_APPLN is used for linking every application with its inventors and applicants. However, this table only builds links to applications who contain **at least a single inventor/applicant with a *Latin* name** (in all available publications of that application). Offices like CN, JP or KR⁴ frequently publish inventor/applicant with the non-Latin names, and only after several months (or never) the transliterated (now Latin) names are additionally made available. Therefore, there would be much more missing location information in PATSTAT, but in the original data in these authorities the issue might be less severe.

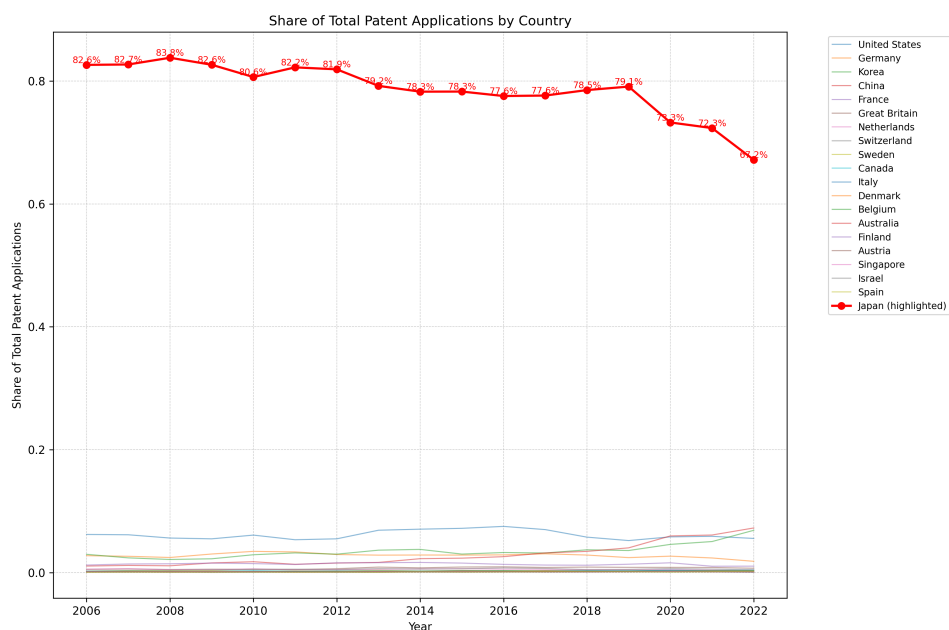
To alleviate this problem, I plan to supplement original data from these two authorities, following de Rassenfosse et al. (2019). I'm currently retrieving data from Chinese patent office.

Since CN and JP accounts for most of LCE inventions, and they both have a large share of source 7 (authority) substitutes, there's no significant differences between baseline method and the adjusted method. However, the concern is I may overestimate the inventions innovated by China and Japan, and also overestimate the importance of China's outbound transfers. I'll check this issue using the original information from Chinese patent office, which is the best information available (Note that Chinese Patent Office only documents applicant locations, not inventor locations). What might alleviate this concern is that seemingly most of the applicants indeed come from the authority country (see fig. 13a and fig. 13b below).

⁴KR has a large share of good quality substitutes, possibly because it has better processing of name translation



(a) Share of Applicant Countries, CN



(b) Share of Applicant Countries, JP

Figure 13: Share of Applicant Countries, 2006-2022

Notes: These figures plot shares of each applicant country in given patent offices (panel (a): CN; panel (b): JP). Note that Chinese Patent Office only documents applicant locations, not inventor locations (only a small number of inventions have this information, before 2010), so applicant countries are actually the best available substitutes for inventor countries. Data Source: PATSNAP.

D Country Income Group

Legend: ■ indicates membership in the income group.

Notes: added in the last column to clarify ambiguous classification/empty values.

Code	Economy	High	Middle	Low	Notes
KE	Kenya		■		
DK	Denmark	■			
OM	Oman	■			
MM	Myanmar		■		
EC	Ecuador		■		
CH	Switzerland	■			
AU	Australia	■			
BN	Brunei Darussalam	■			
FJ	Fiji		■		
BB	Barbados	■			
SC	Seychelles	■			
AT	Austria	■			
PK	Pakistan		■		
GB	United Kingdom	■			
IR	Iran, Islamic Rep.		■		
SV	El Salvador		■		
ES	Spain	■			
VN	Vietnam		■		
AE	United Arab Emirates	■			
AG	Antigua and Barbuda	■			
TR	Turkey		■		
YE	Yemen, Rep.			■	
NG	Nigeria		■		
RO	Romania		■		
UZ	Uzbekistan		■		
GN	Guinea			■	
KN	St. Kitts and Nevis	■			
UY	Uruguay	■			
TW	China		■		
RS	Serbia		■		
DD	Germany	■			
CL	Chile	■			
MU	Mauritius		■		
NA	Namibia		■		
GA	Gabon		■		
HK	China		■		
CN	China		■		
BA	Bosnia and Herzegovina		■		
MA	Morocco		■		
MG	Madagascar			■	

Continued on next page

Code	Economy	High	Middle	Low	Notes
IT	Italy	■			
LR	Liberia			■	
HR	Croatia	■			
TN	Tunisia		■		
SL	Sierra Leone			■	
DZ	Algeria		■		
CA	Canada	■			
BI	Burundi			■	
LV	Latvia	■			
BS	Bahamas, The	■			
CD	Congo, Dem. Rep.			■	
AL	Albania		■		
KR	Korea, Rep.	■			
BH	Bahrain	■			
HU	Hungary	■			
BO	Bolivia		■		
AS	Samoa		■		
JP	Japan	■			
KZ	Kazakhstan		■		
CU	Cuba		■		
SG	Singapore	■			
BG	Bulgaria		■		
PE	Peru		■		
CZ	Czech Republic	■			
ID	Indonesia		■		
IE	Ireland	■			
DJ	Djibouti		■		
SA	Saudi Arabia	■			
KW	Kuwait	■			
ST	São Tomé and Príncipe		■		
AF	Afghanistan			■	
EE	Estonia	■			
IQ	Iraq		■		
TT	Trinidad and Tobago	■			
US	United States	■			
UK	United Kingdom	■			
LY	Libya		■		
BU	Myanmar		■		
ET	Ethiopia			■	
BZ	Belize		■		
MD	Moldova		■		
MK	North Macedonia		■		
GE	Georgia		■		
GH	Ghana		■		
SK	Slovak Republic	■			

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Code	Economy	High	Middle	Low	Notes
HN	Honduras		■		
LA	Lao PDR		■		
LB	Lebanon		■		
CM	Cameroon		■		
ZW	Zimbabwe			■	
FR	France	■			
LU	Luxembourg	■			
IS	Iceland	■			
MO	China		■		
PA	Panama	■			
AN	Netherlands	■			
MN	Mongolia		■		
CY	Cyprus	■			
TO	Tonga		■		
LT	Lithuania	■			
NI	Nicaragua		■		
ML	Mali			■	
UA	Ukraine		■		
TZ	Tanzania			■	
BY	Belarus		■		
AR	Argentina	■			
JO	Jordan		■		
GR	Greece	■			
BE	Belgium	■			
TJ	Tajikistan			■	
BD	Bangladesh		■		
LK	Sri Lanka		■		
BR	Brazil		■		
NE	Niger			■	
ZM	Zambia		■		
AM	Armenia		■		
BJ	Benin			■	
VI	United States	■			
JM	Jamaica		■		
CO	Colombia		■		
TM	Turkmenistan		■		
SN	Senegal			■	
WS	Samoa		■		
IL	Israel	■			
DO	Dominican Republic		■		
MR	Mauritania		■		
MT	Malta	■			
MW	Malawi			■	
MX	Mexico		■		
GY	Guyana		■		

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Code	Economy	High	Middle	Low	Notes
PT	Portugal	■			
EG	Egypt, Arab Rep.		■		
ME	Montenegro		■		
SI	Slovenia	■			
NL	Netherlands	■			
KG	Kyrgyz Republic		■		
FI	Finland	■			
DE	Germany	■			
QA	Qatar	■			
VE	Venezuela, RB		■		
NP	Nepal			■	
PG	Papua New Guinea		■		
PH	Philippines		■		
MY	Malaysia		■		
PL	Poland	■			
SB	Solomon Islands		■		
CS	Montenegro		■		
SZ	Eswatini		■		
AO	Angola		■		
GD	Grenada		■		
ER	Eritrea			■	
BW	Botswana		■		
BT	Bhutan		■		
NO	Norway	■			
RU	Russian Federation		■		
SD	Sudan		■		
SE	Sweden	■			
GT	Guatemala		■		
HT	Haiti			■	
LC	St. Lucia		■		
CG	Congo, Rep.		■		
PY	Paraguay		■		
AZ	Azerbaijan		■		
SR	Suriname		■		
UG	Uganda			■	
SY	Syrian Arab Republic			■	
NZ	New Zealand	■			
CR	Costa Rica		■		
ZA	South Africa		■		
KH	Cambodia		■		
TH	Thailand		■		
TD	Chad			■	
KP	Korea	■			
CI	Côte d'Ivoire		■		
CV	Cabo Verde		■		High Middle

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Code	Economy	High	Middle	Low	Notes
TK	Tokelau				No match
GI	Gibraltar	■			
LI	Liechtenstein	■			
JE	Jersey	■			
IM	Isle of Man	■			
TC	Turks and Caicos Islands	■			
AI	Anguilla	■			
YU	Yugoslavia (historical)				
PR	Puerto Rico (US)	■			
FO	Faroe Islands	■			
AW	Aruba	■			
MH	Marshall Islands	■			High Middle
IO	British Indian Ocean Territory				No match
SM	San Marino	■			
BM	Bermuda	■			
SH	Saint Helena, Ascension and Tristan da Cunha		■		
IN	India		■		Low Middle
CC	Cocos (Keeling) Islands	■			
PS	West Bank and Gaza (Palestine)		■		
GP	Guadeloupe	■			
RE	Réunion	■			
CW	Curaçao	■			
AD	Andorra	■			
KY	Cayman Islands	■			
SX	Sint Maarten (Dutch part)	■			
MC	Monaco	■			
NC	New Caledonia	■			
VG	British Virgin Islands	■			
GL	Greenland	■			
NJ	New Jersey, USA	■			
XX					Not a sovereign economy / unknown (e.g., office/region)
EP		■			
ZZ					Not a sovereign economy / unknown (e.g., office/region)
ON	Ontario, Canada	■			
OA	OAPI			■	Low Middle
NY	New York, USA	■			
AP	ARIPO			■	Low Middle
EA	EAPC		■		High Middle
YS					Not a sovereign economy / unknown (e.g., office/region)
UD					Not a sovereign economy / unknown (e.g., office/region)

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Code	Economy	High	Middle	Low	Notes
CT					Not a sovereign economy / unknown (e.g., office/region)
WO					Not a sovereign economy / unknown (e.g., office/region)
GC		■			
SU					Not a sovereign economy / unknown (e.g., office/region)

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